

HARSHIT CHATURVEDI



IIT Madras
B.S. Degree

9461037924 | 23f1001839@ds.study.iitm.ac.in | linkedin.com/in/harshit-chaturvedi-iitm | github.com/Harshit-Chaturvedi

SUMMARY

Motivated and detail-oriented undergraduate at Indian Institute of Technology, Madras. Experienced in building full-stack applications, machine learning models, and API-based systems through academic and personal projects. Passionate about solving real-world problems using data-driven approaches and continuously learning modern technologies. A quick learner, team player, and adaptable individual with strong analytical and communication skills.

EDUCATION

Indian Institute of Technology
BS in Data Science and Applications
Mansarovar Global University
BSc in Mathematics
Birla Shiksha Kendra
PCM

Chennai, Tamil Nadu
Jan. 2023 – Present
Bhopal, M.P.
Aug. 2020 – May 2023
Chittorgarh, Rajasthan
April 2019 – March 2020

TECHNICAL SKILLS

Languages: Python, Java, SQL, HTML/CSS, JavaScript
Frameworks: Flask, Vue.js, Node.js, Bootstrap, FastAPI, Langchain
GenAI/LLM: LangChain, RAG (Retrieval-Augmented Generation), Vector Databases (Chroma), Prompt Engineering
Developer Tools: Git, GitHub, VS Code, Google Colab, Jupyter Notebook
Libraries: NumPy, Pandas, Matplotlib, Scikit-learn, Seaborn

PROJECTS

- **Cortex — Contextual GenAI Q&A Assistant | Python, LangChain, FastAPI, ChromaDB, Docker**
 - Built a Retrieval-Augmented Generation (RAG) assistant that ingests documents and answers natural-language queries with grounded, cited responses.
 - Implemented document chunking, embedding generation, and semantic search using LangChain and ChromaDB for efficient retrieval.
 - Evaluated retrieval quality using Ragas and optimized the pipeline for improved answer relevance.
 - Developed REST APIs using FastAPI and containerized the application with Docker for deployment.
- **System Threat Forecaster (Kaggle) | Python, Scikit-learn, LightGBM, XGBoost, Pandas**
 - Developed an end-to-end machine learning pipeline to predict malware infection probability using Microsoft's antivirus telemetry dataset.
 - Performed data preprocessing, feature engineering, missing value imputation, encoding, and feature selection on high-dimensional data.
 - Trained and compared LightGBM, XGBoost, Random Forest, and Gradient Boosting models with hyperparameter optimization.
 - Improved prediction performance while reducing false positives through model evaluation and optimization techniques.
- **Household Services Application | Flask, Vue.js, SQLite, Redis, Celery**
 - Developed a full-stack household services platform with role-based dashboards for customers, service professionals, and administrators.
 - Implemented service booking, request tracking, provider management, and secure authentication workflows.
 - Automated reminders and scheduled background tasks using Redis and Celery to improve operational efficiency.
 - Built REST APIs with Flask and an interactive Vue.js frontend for a seamless user experience.
- **Optimizing Seasonal Demand Forecasting and Inventory Management for an Ice-Cream Distributor | Business Data Management Project**
 - Analyzed historical sales data to identify seasonal demand patterns and inventory challenges affecting business performance.
 - Developed a Linear Regression model for demand forecasting using historical sales trends.
 - Recommended inventory optimization strategies to reduce stockouts, overstocking, and product wastage during seasonal fluctuations.
 - Proposed data-driven inventory planning and automated ordering strategies to improve operational efficiency and profitability.

CERTIFICATIONS

- TCS GenAI Powered Data Analytics Job Simulation — Forage
- Mathematical Foundations for Machine Learning — Workshop
- Implementing Machine Learning Techniques using NumPy — Workshop